

Chapter Title: Difference Equations

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Difference Equations

1.1. First-Order Difference Equations

This book is concerned with the dynamic consequences of events over time. Let's say we are studying a variable whose value at date t is denoted y_t . Suppose we are given a dynamic equation relating the value y takes on at date t to another variable w_t and to the value y took on in the previous period:

$$y_t = \phi y_{t-1} + w_t. \quad [1.1.1]$$

Equation [1.1.1] is a *linear first-order difference equation*. A *difference equation* is an expression relating a variable y_t to its previous values. This is a *first-order* difference equation because only the first lag of the variable (y_{t-1}) appears in the equation. Note that it expresses y_t as a linear function of y_{t-1} and w_t .

An example of [1.1.1] is Goldfeld's (1973) estimated money demand function for the United States. Goldfeld's model related the log of the real money holdings of the public (m_t) to the log of aggregate real income (I_t), the log of the interest rate on bank accounts (r_{bt}), and the log of the interest rate on commercial paper (r_{ct}):

$$m_t = 0.27 + 0.72m_{t-1} + 0.19I_t - 0.045r_{bt} - 0.019r_{ct}. \quad [1.1.2]$$

This is a special case of [1.1.1] with $y_t = m_t$, $\phi = 0.72$, and

$$w_t = 0.27 + 0.19I_t - 0.045r_{bt} - 0.019r_{ct}.$$

For purposes of analyzing the dynamics of such a system, it simplifies the algebra a little to summarize the effects of all the input variables (I_t , r_{bt} , and r_{ct}) in terms of a scalar w_t as here.

In Chapter 3 the input variable w_t will be regarded as a random variable, and the implications of [1.1.1] for the statistical properties of the output series y_t will be explored. In preparation for this discussion, it is necessary first to understand the mechanics of difference equations. For the discussion in Chapters 1 and 2, the values for the input variable $\{w_1, w_2, \dots\}$ will simply be regarded as a sequence of deterministic numbers. Our goal is to answer the following question: If a dynamic system is described by [1.1.1], what are the effects on y of changes in the value of w ?

Solving a Difference Equation by Recursive Substitution

The presumption is that the dynamic equation [1.1.1] governs the behavior of y for all dates t . Thus, for each date we have an equation relating the value of

y for that date to its previous value and the current value of w :

<i>Date</i>	<i>Equation</i>	
0	$y_0 = \phi y_{-1} + w_0$	[1.1.3]
1	$y_1 = \phi y_0 + w_1$	[1.1.4]
2	$y_2 = \phi y_1 + w_2$	[1.1.5]
$\vdots$	$\vdots$	
t	$y_t = \phi y_{t-1} + w_t$	[1.1.6]

If we know the starting value of y for date $t = -1$ and the value of w for dates $t = 0, 1, 2, \dots$, then it is possible to simulate this dynamic system to find the value of y for any date. For example, if we know the value of y for $t = -1$ and the value of w for $t = 0$, we can calculate the value of y for $t = 0$ directly from [1.1.3]. Given this value of y_0 and the value of w for $t = 1$, we can calculate the value of y for $t = 1$ from [1.1.4]:

$$y_1 = \phi y_0 + w_1 = \phi(\phi y_{-1} + w_0) + w_1,$$

or

$$y_1 = \phi^2 y_{-1} + \phi w_0 + w_1.$$

Given this value of y_1 and the value of w for $t = 2$, we can calculate the value of y for $t = 2$ from [1.1.5]:

$$y_2 = \phi y_1 + w_2 = \phi(\phi^2 y_{-1} + \phi w_0 + w_1) + w_2,$$

or

$$y_2 = \phi^3 y_{-1} + \phi^2 w_0 + \phi w_1 + w_2.$$

Continuing recursively in this fashion, the value that y takes on at date t can be described as a function of its initial value y_{-1} and the history of w between date 0 and date t :

$$y_t = \phi^{t+1} y_{-1} + \phi^t w_0 + \phi^{t-1} w_1 + \phi^{t-2} w_2 + \dots + \phi w_{t-1} + w_t. \quad [1.1.7]$$

This procedure is known as solving the difference equation [1.1.1] by *recursive substitution*.

Dynamic Multipliers

Note that [1.1.7] expresses y_t as a linear function of the initial value y_{-1} and the historical values of w . This makes it very easy to calculate the effect of w_0 on y_t . If w_0 were to change with y_{-1} and $w_1, w_2, \dots, w_t$ taken as unaffected, the effect on y_t would be given by

$$\frac{\partial y_t}{\partial w_0} = \phi^t. \quad [1.1.8]$$

Note that the calculations would be exactly the same if the dynamic simulation were started at date t (taking y_{t-1} as given); then y_{t+j} could be described as a

function of y_{t-1} and $w_t, w_{t+1}, \dots, w_{t+j}$:

$$y_{t+j} = \phi^{j+1}y_{t-1} + \phi^j w_t + \phi^{j-1}w_{t+1} + \phi^{j-2}w_{t+2} + \dots + \phi w_{t+j-1} + w_{t+j}. \quad [1.1.9]$$

The effect of w_t on y_{t+j} is given by

$$\frac{\partial y_{t+j}}{\partial w_t} = \phi^j. \quad [1.1.10]$$

Thus the *dynamic multiplier* [1.1.10] depends only on j , the length of time separating the disturbance to the input (w_t) and the observed value of the output (y_{t+j}). The multiplier does not depend on t ; that is, it does not depend on the dates of the observations themselves. This is true of any linear difference equation.

As an example of calculating a dynamic multiplier, consider again Goldfeld's money demand specification [1.1.2]. Suppose we want to know what will happen to money demand two quarters from now if current income I_t were to increase by one unit today with future income I_{t+1} and I_{t+2} unaffected:

$$\frac{\partial m_{t+2}}{\partial I_t} = \frac{\partial m_{t+2}}{\partial w_t} \times \frac{\partial w_t}{\partial I_t} = \phi^2 \times \frac{\partial w_t}{\partial I_t}.$$

From [1.1.2], a one-unit increase in I_t will increase w_t by 0.19 units, meaning that $\partial w_t / \partial I_t = 0.19$. Since $\phi = 0.72$, we calculate

$$\frac{\partial m_{t+2}}{\partial I_t} = (0.72)^2(0.19) = 0.098.$$

Because I_t is the log of income, an increase in I_t of 0.01 units corresponds to a 1% increase in income. An increase in m_t of $(0.01) \cdot (0.098) \cong 0.001$ corresponds to a 0.1% increase in money holdings. Thus the public would be expected to increase its money holdings by a little less than 0.1% two quarters following a 1% increase in income.

Different values of ϕ in [1.1.1] can produce a variety of dynamic responses of y to w . If $0 < \phi < 1$, the multiplier $\partial y_{t+j} / \partial w_t$ in [1.1.10] decays geometrically toward zero. Panel (a) of Figure 1.1 plots ϕ^j as a function of j for $\phi = 0.8$. If $-1 < \phi < 0$, the multiplier $\partial y_{t+j} / \partial w_t$ will alternate in sign as in panel (b). In this case an increase in w_t will cause y_t to be higher, y_{t+1} to be lower, y_{t+2} to be higher, and so on. Again the absolute value of the effect decays geometrically toward zero. If $\phi > 1$, the dynamic multiplier increases exponentially over time as in panel (c). A given increase in w_t has a larger effect the farther into the future one goes. For $\phi < -1$, the system [1.1.1] exhibits explosive oscillation as in panel (d).

Thus, if $|\phi| < 1$, the system is stable; the consequences of a given change in w_t will eventually die out. If $|\phi| > 1$, the system is explosive. An interesting possibility is the borderline case, $\phi = 1$. In this case, the solution [1.1.9] becomes

$$y_{t+j} = y_{t-1} + w_t + w_{t+1} + w_{t+2} + \dots + w_{t+j-1} + w_{t+j}. \quad [1.1.11]$$

Here the output variable y is the sum of the historical inputs w . A one-unit increase in w will cause a permanent one-unit increase in y :

$$\frac{\partial y_{t+j}}{\partial w_t} = 1 \quad \text{for } j = 0, 1, \dots$$

We might also be interested in the effect of w on the present value of the stream of future realizations of y . For a given stream of future values $y_t, y_{t+1},$

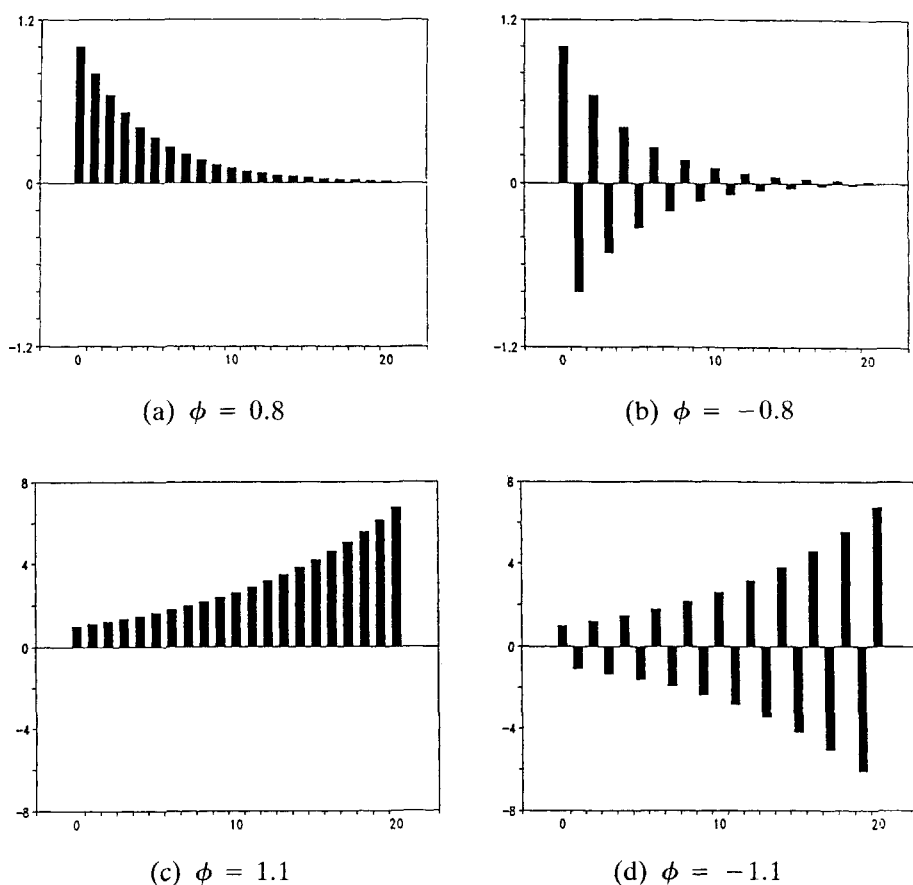


FIGURE 1.1 Dynamic multiplier for first-order difference equation for different values of ϕ (plot of $\partial y_{t+j}/\partial w_t = \phi^j$ as a function of the lag j).

$y_{t+2}, \dots$ and a constant interest rate¹ $r > 0$, the *present value* of the stream at time t is given by

$$y_t + \frac{y_{t+1}}{1+r} + \frac{y_{t+2}}{(1+r)^2} + \frac{y_{t+3}}{(1+r)^3} + \dots \quad [1.1.12]$$

Let β denote the discount factor:

$$\beta \equiv 1/(1+r).$$

Note that $0 < \beta < 1$. Then the present value [1.1.12] can be written as

$$\sum_{j=0}^{\infty} \beta^j y_{t+j}. \quad [1.1.13]$$

Consider what would happen if there were a one-unit increase in w_t with $w_{t+1}, w_{t+2}, \dots$ unaffected. The consequences of this change for the present value of y are found by differentiating [1.1.13] with respect to w_t and then using [1.1.10]

¹The interest rate is measured here as a fraction of 1; thus $r = 0.1$ corresponds to a 10% interest rate.

to evaluate each derivative:

$$\sum_{j=0}^{\infty} \beta^j \frac{\partial y_{t+j}}{\partial w_t} = \sum_{j=0}^{\infty} \beta^j \phi^j = 1/(1 - \beta\phi), \quad [1.1.14]$$

provided that $|\beta\phi| < 1$.

In calculating the dynamic multipliers [1.1.10] or [1.1.14], we were asking what would happen if w_t were to increase by one unit with $w_{t+1}, w_{t+2}, \dots, w_{t+j}$ unaffected. We were thus finding the effect of a purely transitory change in w . Panel (a) of Figure 1.2 shows the time path of w associated with this question, and panel (b) shows the implied path for y . Because the dynamic multiplier [1.1.10] calculates the response of y to a single impulse in w , it is also referred to as the *impulse-response function*.

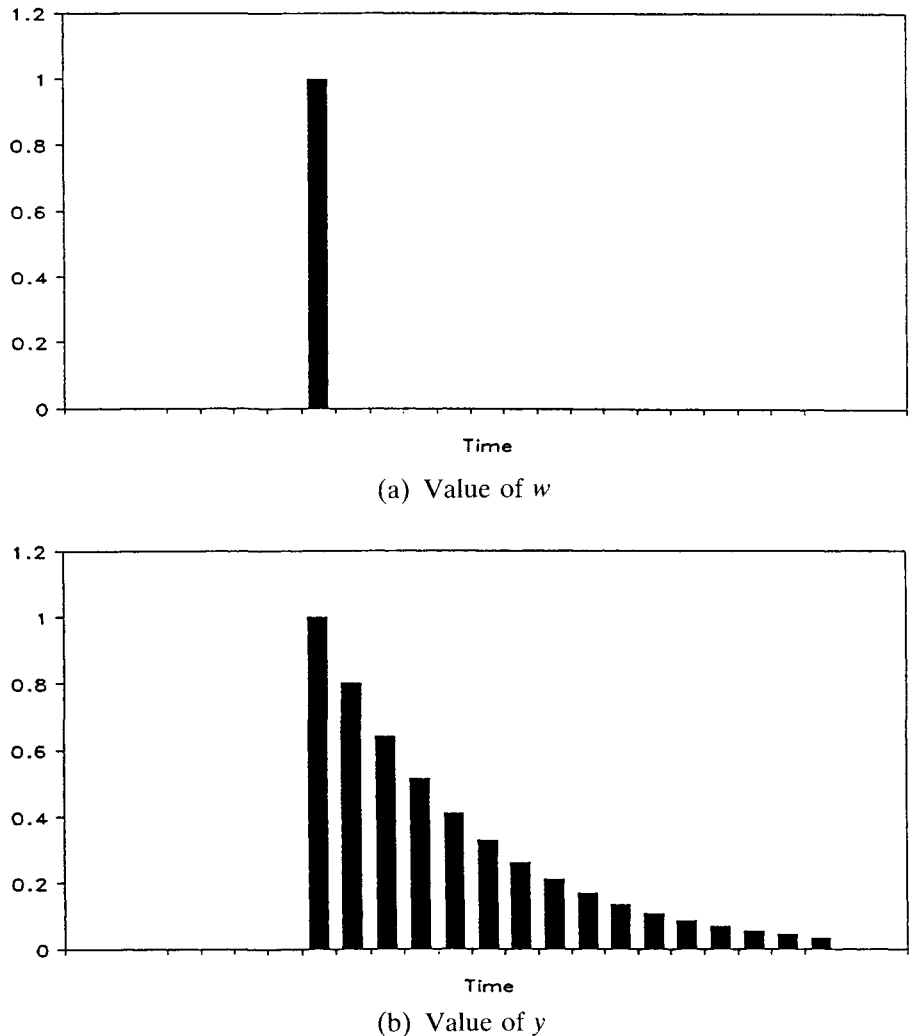


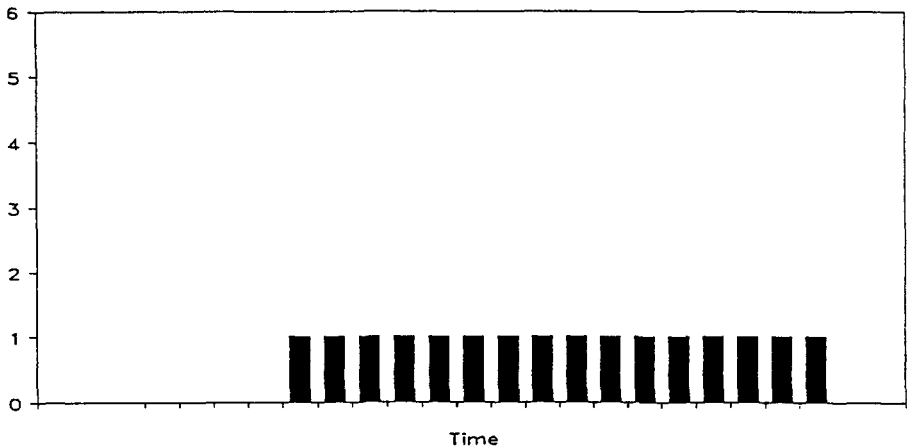
FIGURE 1.2 Paths of input variable (w_t) and output variable (y_t) assumed for dynamic multiplier and present-value calculations.

Sometimes we might instead be interested in the consequences of a permanent change in w . A permanent change in w means that $w_t, w_{t+1}, \dots$, and w_{t+j} would all increase by one unit, as in Figure 1.3. From formula [1.1.10], the effect on y_{t+j} of a permanent change in w beginning in period t is given by

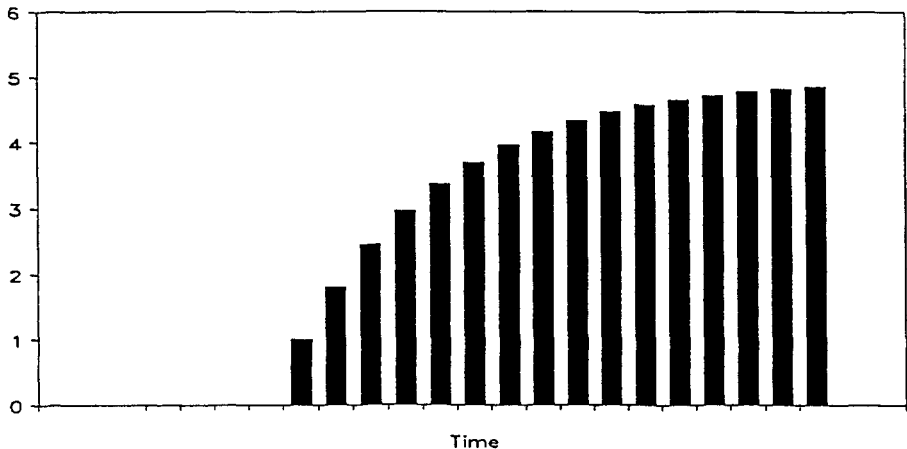
$$\frac{\partial y_{t+j}}{\partial w_t} + \frac{\partial y_{t+j}}{\partial w_{t+1}} + \frac{\partial y_{t+j}}{\partial w_{t+2}} + \dots + \frac{\partial y_{t+j}}{\partial w_{t+j}} = \phi^j + \phi^{j-1} + \phi^{j-2} + \dots + \phi + 1.$$

When $|\phi| < 1$, the limit of this expression as j goes to infinity is sometimes described as the “long-run” effect of w on y :

$$\lim_{j \rightarrow \infty} \left[\frac{\partial y_{t+j}}{\partial w_t} + \frac{\partial y_{t+j}}{\partial w_{t+1}} + \frac{\partial y_{t+j}}{\partial w_{t+2}} + \dots + \frac{\partial y_{t+j}}{\partial w_{t+j}} \right] = 1 + \phi + \phi^2 + \dots = 1/(1 - \phi). \quad [1.1.15]$$



(a) Value of w



(b) Value of y

FIGURE 1.3 Paths of input variable (w_t) and output variable (y_t) assumed for long-run effect calculations.

For example, the long-run income elasticity of money demand in the system [1.1.2] is given by

$$\frac{0.19}{1 - 0.72} = 0.68.$$

A permanent 1% increase in income will eventually lead to a 0.68% increase in money demand.

Another related question concerns the cumulative consequences for y of a one-time change in w . Here we consider a transitory disturbance to w as in panel (a) of Figure 1.2, but wish to calculate the sum of the consequences for all future values of y . Another way to think of this is as the effect on the present value of y [1.1.13] with the discount rate $\beta = 1$. Setting $\beta = 1$ in [1.1.14] shows this cumulative effect to be equal to

$$\sum_{j=0}^{\infty} \frac{\partial y_{t+j}}{\partial w_t} = 1/(1 - \phi), \quad [1.1.16]$$

provided that $|\phi| < 1$. Note that the cumulative effect on y of a transitory change in w (expression [1.1.16]) is the same as the long-run effect on y of a permanent change in w (expression [1.1.15]).

1.2. *pth-Order Difference Equations*

Let us now generalize the dynamic system [1.1.1] by allowing the value of y at date t to depend on p of its own lags along with the current value of the input variable w_t :

$$y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \cdots + \phi_p y_{t-p} + w_t. \quad [1.2.1]$$

Equation [1.2.1] is a linear p th-order difference equation.

It is often convenient to rewrite the p th-order difference equation [1.2.1] in the scalar y_t as a first-order difference equation in a vector ξ_t . Define the $(p \times 1)$ vector ξ_t by

$$\xi_t \equiv \begin{bmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p+1} \end{bmatrix}. \quad [1.2.2]$$

That is, the first element of the vector ξ at date t is the value y took on at date t . The second element of ξ_t is the value y took on at date $t - 1$, and so on. Define the $(p \times p)$ matrix F by

$$F \equiv \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix}. \quad [1.2.3]$$

For example, for $p = 4$, $\mathbf{F}$ refers to the following 4×4 matrix:

$$\mathbf{F} = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \phi_4 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}.$$

For $p = 1$ (the first-order difference equation [1.1.1]), $\mathbf{F}$ is just the scalar ϕ . Finally, define the $(p \times 1)$ vector $\mathbf{v}_t$ by

$$\mathbf{v}_t \equiv \begin{bmatrix} w_t \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \quad [1.2.4]$$

Consider the following first-order vector difference equation:

$$\xi_t = \mathbf{F}\xi_{t-1} + \mathbf{v}_t, \quad [1.2.5]$$

or

$$\begin{bmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p+1} \end{bmatrix} = \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \dots & \phi_{p-1} & \phi_p \\ 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 1 & 0 \end{bmatrix} \begin{bmatrix} y_{t-1} \\ y_{t-2} \\ y_{t-3} \\ \vdots \\ y_{t-p} \end{bmatrix} + \begin{bmatrix} w_t \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

This is a system of p equations. The first equation in this system is identical to equation [1.2.1]. The second equation is simply the identity

$$y_{t-1} = y_{t-1},$$

owing to the fact that the second element of ξ_t is the same as the first element of ξ_{t-1} . The third equation in [1.2.5] states that $y_{t-2} = y_{t-2}$; the p th equation states that $y_{t-p+1} = y_{t-p+1}$.

Thus, the first-order vector system [1.2.5] is simply an alternative representation of the p th-order scalar system [1.2.1]. The advantage of rewriting the p th-order system [1.2.1] in the form of a first-order system [1.2.5] is that first-order systems are often easier to work with than p th-order systems.

A dynamic multiplier for [1.2.5] can be found in exactly the same way as was done for the first-order scalar system of Section 1.1. If we knew the value of the vector ξ for date $t = -1$ and of $\mathbf{v}$ for date $t = 0$, we could find the value of ξ for date 0 from

$$\xi_0 = \mathbf{F}\xi_{-1} + \mathbf{v}_0.$$

The value of ξ for date 1 is

$$\xi_1 = \mathbf{F}\xi_0 + \mathbf{v}_1 = \mathbf{F}(\mathbf{F}\xi_{-1} + \mathbf{v}_0) + \mathbf{v}_1 = \mathbf{F}^2\xi_{-1} + \mathbf{F}\mathbf{v}_0 + \mathbf{v}_1.$$

Proceeding recursively in this fashion produces a generalization of [1.1.7]:

$$\xi_t = \mathbf{F}^{t+1}\xi_{-1} + \mathbf{F}^t\mathbf{v}_0 + \mathbf{F}^{t-1}\mathbf{v}_1 + \mathbf{F}^{t-2}\mathbf{v}_2 + \dots + \mathbf{F}\mathbf{v}_{t-1} + \mathbf{v}_t. \quad [1.2.6]$$

Writing this out in terms of the definitions of ξ and $\mathbf{v}$,

$$\begin{aligned} \begin{bmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ \vdots \\ y_{t-p+1} \end{bmatrix} &= \mathbf{F}^{t+1} \begin{bmatrix} y_{-1} \\ y_{-2} \\ y_{-3} \\ \vdots \\ y_{-p} \end{bmatrix} + \mathbf{F}^t \begin{bmatrix} w_0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \mathbf{F}^{t-1} \begin{bmatrix} w_1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \cdots \\ &\quad + \mathbf{F}^1 \begin{bmatrix} w_{t-1} \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + \begin{bmatrix} w_t \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}. \end{aligned} \quad [1.2.7]$$

Consider the first equation of this system, which characterizes the value of y_t . Let $f_{11}^{(t)}$ denote the (1, 1) element of $\mathbf{F}^t$, $f_{12}^{(t)}$ the (1, 2) element of $\mathbf{F}^t$, and so on. Then the first equation of [1.2.7] states that

$$\begin{aligned} y_t &= f_{11}^{(t+1)}y_{-1} + f_{12}^{(t+1)}y_{-2} + \cdots + f_{1p}^{(t+1)}y_{-p} + f_{11}^{(t)}w_0 \\ &\quad + f_{11}^{(t-1)}w_1 + \cdots + f_{11}^{(1)}w_{t-1} + w_t. \end{aligned} \quad [1.2.8]$$

This describes the value of y at date t as a linear function of p initial values of y ($y_{-1}, y_{-2}, \dots, y_{-p}$) and the history of the input variable w since time 0 ($w_0, w_1, \dots, w_t$). Note that whereas only one initial value for y (the value y_{-1}) was needed in the case of a first-order difference equation, p initial values for y (the values $y_{-1}, y_{-2}, \dots, y_{-p}$) are needed in the case of a p th-order difference equation.

The obvious generalization of [1.1.9] is

$$\begin{aligned} \xi_{t+j} &= \mathbf{F}^{j+1}\xi_{t-1} + \mathbf{F}^j\mathbf{v}_t + \mathbf{F}^{j-1}\mathbf{v}_{t+1} + \mathbf{F}^{j-2}\mathbf{v}_{t+2} + \cdots \\ &\quad + \mathbf{F}\mathbf{v}_{t+j-1} + \mathbf{v}_{t+j} \end{aligned} \quad [1.2.9]$$

from which

$$\begin{aligned} y_{t+j} &= f_{11}^{(j+1)}y_{t-1} + f_{12}^{(j+1)}y_{t-2} + \cdots + f_{1p}^{(j+1)}y_{t-p} + f_{11}^{(j)}w_t \\ &\quad + f_{11}^{(j-1)}w_{t+1} + f_{11}^{(j-2)}w_{t+2} + \cdots + f_{11}^{(1)}w_{t+j-1} + w_{t+j}. \end{aligned} \quad [1.2.10]$$

Thus, for a p th-order difference equation, the dynamic multiplier is given by

$$\frac{\partial y_{t+j}}{\partial w_t} = f_{11}^{(j)} \quad [1.2.11]$$

where $f_{11}^{(j)}$ denotes the (1, 1) element of $\mathbf{F}^j$. For $j = 1$, this is simply the (1, 1) element of $\mathbf{F}$, or the parameter ϕ_1 . Thus, for any p th-order system, the effect on y_{t+1} of a one-unit increase in w_t is given by the coefficient relating y_t to y_{t-1} in equation [1.2.1]:

$$\frac{\partial y_{t+1}}{\partial w_t} = \phi_1.$$

Direct multiplication of [1.2.3] reveals that the $(1, 1)$ element of $\mathbf{F}^2$ is $(\phi_1^2 + \phi_2)$, so

$$\frac{\partial y_{t+2}}{\partial w_t} = \phi_1^2 + \phi_2$$

in a p th-order system.

For larger values of j , an easy way to obtain a numerical value for the dynamic multiplier $\partial y_{t+j}/\partial w_t$ is to simulate the system. This is done as follows. Set $y_{-1} = y_{-2} = \dots = y_{-p} = 0$, $w_0 = 1$, and set the value of w for all other dates to 0. Then use [1.2.1] to calculate the value of y_t for $t = 0$ (namely, $y_0 = 1$). Next substitute this value along with $y_{t-1}, y_{t-2}, \dots, y_{t-p+1}$ back into [1.2.1] to calculate y_{t+1} , and continue recursively in this fashion. The value of y at step t gives the effect of a one-unit change in w_0 on y_t .

Although numerical simulation may be adequate for many circumstances, it is also useful to have a simple analytical characterization of $\partial y_{t+j}/\partial w_t$, which, we know from [1.2.11], is given by the $(1, 1)$ element of $\mathbf{F}^j$. This is fairly easy to obtain in terms of the eigenvalues of the matrix $\mathbf{F}$. Recall that the eigenvalues of a matrix $\mathbf{F}$ are those numbers λ for which

$$|\mathbf{F} - \lambda \mathbf{I}_p| = 0. \quad [1.2.12]$$

For example, for $p = 2$ the eigenvalues are the solutions to

$$\left| \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

or

$$\begin{vmatrix} (\phi_1 - \lambda) & \phi_2 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \phi_1\lambda - \phi_2 = 0. \quad [1.2.13]$$

The two eigenvalues of $\mathbf{F}$ for a second-order difference equation are thus given by

$$\lambda_1 = \frac{\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}}{2} \quad [1.2.14]$$

$$\lambda_2 = \frac{\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2}. \quad [1.2.15]$$

For a general p th-order system, the determinant in [1.2.12] is a p th-order polynomial in λ whose p solutions characterize the p eigenvalues of $\mathbf{F}$. This polynomial turns out to take a very similar form to [1.2.13]. The following result is proved in Appendix 1.A at the end of this chapter.

Proposition 1.1: *The eigenvalues of the matrix $\mathbf{F}$ defined in equation [1.2.3] are the values of λ that satisfy*

$$\lambda^p - \phi_1\lambda^{p-1} - \phi_2\lambda^{p-2} - \dots - \phi_{p-1}\lambda - \phi_p = 0. \quad [1.2.16]$$

Once we know the eigenvalues, it is straightforward to characterize the dynamic behavior of the system. First we consider the case when the eigenvalues of $\mathbf{F}$ are distinct; for example, we require that λ_1 and λ_2 in [1.2.14] and [1.2.15] be different numbers.

General Solution of a p th-Order Difference Equation with Distinct Eigenvalues

Recall² that if the eigenvalues of a $(p \times p)$ matrix $\mathbf{F}$ are distinct, there exists a nonsingular $(p \times p)$ matrix $\mathbf{T}$ such that

$$\mathbf{F} = \mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1} \quad [1.2.17]$$

where $\mathbf{\Lambda}$ is a $(p \times p)$ matrix with the eigenvalues of $\mathbf{F}$ along the principal diagonal and zeros elsewhere:

$$\mathbf{\Lambda} = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_p \end{bmatrix}. \quad [1.2.18]$$

This enables us to characterize the dynamic multiplier (the $(1, 1)$ element of $\mathbf{F}^j$ in [1.2.11]) very easily. For example, from [1.2.17] we can write $\mathbf{F}^2$ as

$$\begin{aligned} \mathbf{F}^2 &= \mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1} \times \mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1} \\ &= \mathbf{T} \times \mathbf{\Lambda} \times (\mathbf{T}^{-1}\mathbf{T}) \times \mathbf{\Lambda} \times \mathbf{T}^{-1} \\ &= \mathbf{T} \times \mathbf{\Lambda} \times \mathbf{I}_p \times \mathbf{\Lambda} \times \mathbf{T}^{-1} \\ &= \mathbf{T}\mathbf{\Lambda}^2\mathbf{T}^{-1}. \end{aligned}$$

The diagonal structure of $\mathbf{\Lambda}$ implies that $\mathbf{\Lambda}^2$ is also a diagonal matrix whose elements are the squares of the eigenvalues of $\mathbf{F}$:

$$\mathbf{\Lambda}^2 = \begin{bmatrix} \lambda_1^2 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2^2 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_p^2 \end{bmatrix}.$$

More generally, we can characterize $\mathbf{F}^j$ in terms of the eigenvalues of $\mathbf{F}$ as

$$\begin{aligned} \mathbf{F}^j &= \underbrace{\mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1} \times \mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1} \times \cdots \times \mathbf{T}\mathbf{\Lambda}\mathbf{T}^{-1}}_{j \text{ terms}} \\ &= \mathbf{T} \times \mathbf{\Lambda} \times (\mathbf{T}^{-1}\mathbf{T}) \times \mathbf{\Lambda} \times (\mathbf{T}^{-1}\mathbf{T}) \times \cdots \times \mathbf{\Lambda} \times \mathbf{T}^{-1}, \end{aligned}$$

which simplifies to

$$\mathbf{F}^j = \mathbf{T}\mathbf{\Lambda}^j\mathbf{T}^{-1} \quad [1.2.19]$$

where

$$\mathbf{\Lambda}^j = \begin{bmatrix} \lambda_1^j & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2^j & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_p^j \end{bmatrix}.$$

²See equation [A.4.24] in the Mathematical Review (Appendix A) at the end of the book.

Let t_{ij} denote the row i , column j element of $\mathbf{T}$ and let t^{ij} denote the row i , column j element of $\mathbf{T}^{-1}$. Equation [1.2.19] written out explicitly becomes

$$\begin{aligned}\mathbf{F}^j &= \begin{bmatrix} t_{11} & t_{12} & \cdots & t_{1p} \\ t_{21} & t_{22} & \cdots & t_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ t_{p1} & t_{p2} & \cdots & t_{pp} \end{bmatrix} \begin{bmatrix} \lambda_1^j & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2^j & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_p^j \end{bmatrix} \begin{bmatrix} t^{11} & t^{12} & \cdots & t^{1p} \\ t^{21} & t^{22} & \cdots & t^{2p} \\ \vdots & \vdots & \cdots & \vdots \\ t^{p1} & t^{p2} & \cdots & t^{pp} \end{bmatrix} \\ &= \begin{bmatrix} t_{11}\lambda_1^j & t_{12}\lambda_2^j & \cdots & t_{1p}\lambda_p^j \\ t_{21}\lambda_1^j & t_{22}\lambda_2^j & \cdots & t_{2p}\lambda_p^j \\ \vdots & \vdots & \cdots & \vdots \\ t_{p1}\lambda_1^j & t_{p2}\lambda_2^j & \cdots & t_{pp}\lambda_p^j \end{bmatrix} \begin{bmatrix} t^{11} & t^{12} & \cdots & t^{1p} \\ t^{21} & t^{22} & \cdots & t^{2p} \\ \vdots & \vdots & \cdots & \vdots \\ t^{p1} & t^{p2} & \cdots & t^{pp} \end{bmatrix}\end{aligned}$$

from which the $(1, 1)$ element of $\mathbf{F}^j$ is given by

$$f_{11}^{(j)} = [t_{11}t^{11}]\lambda_1^j + [t_{12}t^{21}]\lambda_2^j + \cdots + [t_{1p}t^{p1}]\lambda_p^j$$

or

$$f_{11}^{(j)} = c_1\lambda_1^j + c_2\lambda_2^j + \cdots + c_p\lambda_p^j \quad [1.2.20]$$

where

$$c_i = [t_{1i}t^{i1}]. \quad [1.2.21]$$

Note that the sum of the c_i terms has the following interpretation:

$$c_1 + c_2 + \cdots + c_p = [t_{11}t^{11}] + [t_{12}t^{21}] + \cdots + [t_{1p}t^{p1}], \quad [1.2.22]$$

which is the $(1, 1)$ element of $\mathbf{T} \cdot \mathbf{T}^{-1}$. Since $\mathbf{T} \cdot \mathbf{T}^{-1}$ is just the $(p \times p)$ identity matrix, [1.2.22] implies that the c_i terms sum to unity:

$$c_1 + c_2 + \cdots + c_p = 1. \quad [1.2.23]$$

Substituting [1.2.20] into [1.2.11] gives the form of the dynamic multiplier for a p th-order difference equation:

$$\frac{\partial y_{t+j}}{\partial w_t} = c_1\lambda_1^j + c_2\lambda_2^j + \cdots + c_p\lambda_p^j. \quad [1.2.24]$$

Equation [1.2.24] characterizes the dynamic multiplier as a weighted average of each of the p eigenvalues raised to the j th power.

The following result provides a closed-form expression for the constants $(c_1, c_2, \dots, c_p)$.

Proposition 1.2: *If the eigenvalues $(\lambda_1, \lambda_2, \dots, \lambda_p)$ of the matrix $\mathbf{F}$ in [1.2.3] are distinct, then the magnitude c_i in [1.2.21] can be written*

$$c_i = \frac{\lambda_i^{p-1}}{\prod_{\substack{k=1 \\ k \neq i}}^p (\lambda_i - \lambda_k)}. \quad [1.2.25]$$

To summarize, the p th-order difference equation [1.2.1] implies that

$$\begin{aligned}y_{t+j} &= f_{11}^{(j+1)}y_{t-1} + f_{12}^{(j+1)}y_{t-2} + \cdots + f_{1p}^{(j+1)}y_{t-p} \\ &\quad + w_{t+j} + \psi_1w_{t+j-1} + \psi_2w_{t+j-2} + \cdots + \psi_{j-1}w_{t+1} + \psi_jw_t.\end{aligned} \quad [1.2.26]$$

The dynamic multiplier

$$\frac{\partial y_{t+j}}{\partial w_t} = \psi_j \quad [1.2.27]$$

is given by the (1, 1) element of $\mathbf{F}^j$:

$$\psi_j = f_{11}^{(j)}. \quad [1.2.28]$$

A closed-form expression for ψ_j can be obtained by finding the eigenvalues of $\mathbf{F}$, or the values of λ satisfying [1.2.16]. Denoting these p values by $(\lambda_1, \lambda_2, \dots, \lambda_p)$ and assuming them to be distinct, the dynamic multiplier is given by

$$\psi_j = c_1 \lambda_1^j + c_2 \lambda_2^j + \dots + c_p \lambda_p^j \quad [1.2.29]$$

where $(c_1, c_2, \dots, c_p)$ is a set of constants summing to unity given by expression [1.2.25].

For a first-order system ($p = 1$), this rule would have us solve [1.2.16],

$$\lambda - \phi_1 = 0,$$

which has the single solution

$$\lambda_1 = \phi_1. \quad [1.2.30]$$

According to [1.2.29], the dynamic multiplier is given by

$$\frac{\partial y_{t+j}}{\partial w_t} = c_1 \lambda_1^j. \quad [1.2.31]$$

From [1.2.23], $c_1 = 1$. Substituting this and [1.2.30] into [1.2.31] gives

$$\frac{\partial y_{t+j}}{\partial w_t} = \phi_1^j,$$

or the same result found in Section 1.1.

For higher-order systems, [1.2.29] allows a variety of more complicated dynamics. Suppose first that all the eigenvalues of $\mathbf{F}$ (or solutions to [1.2.16]) are real. This would be the case, for example, if $p = 2$ and $\phi_1^2 + 4\phi_2 > 0$ in the solutions [1.2.14] and [1.2.15] for the second-order system. If, furthermore, all of the eigenvalues are less than 1 in absolute value, then the system is stable, and its dynamics are represented as a weighted average of decaying exponentials or decaying exponentials oscillating in sign. For example, consider the following second-order difference equation:

$$y_t = 0.6y_{t-1} + 0.2y_{t-2} + w_t.$$

From equations [1.2.14] and [1.2.15], the eigenvalues of this system are given by

$$\begin{aligned} \lambda_1 &= \frac{0.6 + \sqrt{(0.6)^2 + 4(0.2)}}{2} = 0.84 \\ \lambda_2 &= \frac{0.6 - \sqrt{(0.6)^2 + 4(0.2)}}{2} = -0.24. \end{aligned}$$

From [1.2.25], we have

$$\begin{aligned} c_1 &= \lambda_1/(\lambda_1 - \lambda_2) = 0.778 \\ c_2 &= \lambda_2/(\lambda_2 - \lambda_1) = 0.222. \end{aligned}$$

The dynamic multiplier for this system,

$$\frac{\partial y_{t+j}}{\partial w_t} = c_1 \lambda_1^j + c_2 \lambda_2^j,$$

is plotted as a function of j in panel (a) of Figure 1.4.³ Note that as j becomes larger, the pattern is dominated by the larger eigenvalue (λ_1), approximating a simple geometric decay at rate λ_1 .

If the eigenvalues (the solutions to [1.2.16]) are real but at least one is greater than unity in absolute value, the system is explosive. If λ_1 denotes the eigenvalue that is largest in absolute value, the dynamic multiplier is eventually dominated by an exponential function of that eigenvalue:

$$\lim_{j \rightarrow \infty} \frac{\partial y_{t+j}}{\partial w_t} \cdot \frac{1}{\lambda_1^j} = c_1.$$

Other interesting possibilities arise if some of the eigenvalues are complex. Whenever this is the case, they appear as complex conjugates. For example, if $p = 2$ and $\phi_1^2 + 4\phi_2 < 0$, then the solutions λ_1 and λ_2 in [1.2.14] and [1.2.15] are complex conjugates. Suppose that λ_1 and λ_2 are complex conjugates, written as

$$\lambda_1 = a + bi \quad [1.2.32]$$

$$\lambda_2 = a - bi. \quad [1.2.33]$$

For the $p = 2$ case of [1.2.14] and [1.2.15], we would have

$$a = \phi_1/2 \quad [1.2.34]$$

$$b = (1/2)\sqrt{-\phi_1^2 - 4\phi_2}. \quad [1.2.35]$$

Our goal is to characterize the contribution to the dynamic multiplier $c_1\lambda_1^j$ when λ_1 is a complex number as in [1.2.32]. Recall that to raise a complex number to a power, we rewrite [1.2.32] in polar coordinate form:

$$\lambda_1 = R[\cos(\theta) + i\sin(\theta)], \quad [1.2.36]$$

where θ and R are defined in terms of a and b by the following equations:

$$R = \sqrt{a^2 + b^2}$$

$$\cos(\theta) = a/R$$

$$\sin(\theta) = b/R.$$

Note that R is equal to the modulus of the complex number λ_1 .

The eigenvalue λ_1 in [1.2.36] can be written as⁴

$$\lambda_1 = R[e^{i\theta}],$$

and so

$$\lambda_1^j = R^j[e^{i\theta j}] = R^j[\cos(\theta j) + i\sin(\theta j)]. \quad [1.2.37]$$

Analogously, if λ_2 is the complex conjugate of λ_1 , then

$$\lambda_2 = R[\cos(\theta) - i\sin(\theta)],$$

which can be written⁵

$$\lambda_2 = R[e^{-i\theta}].$$

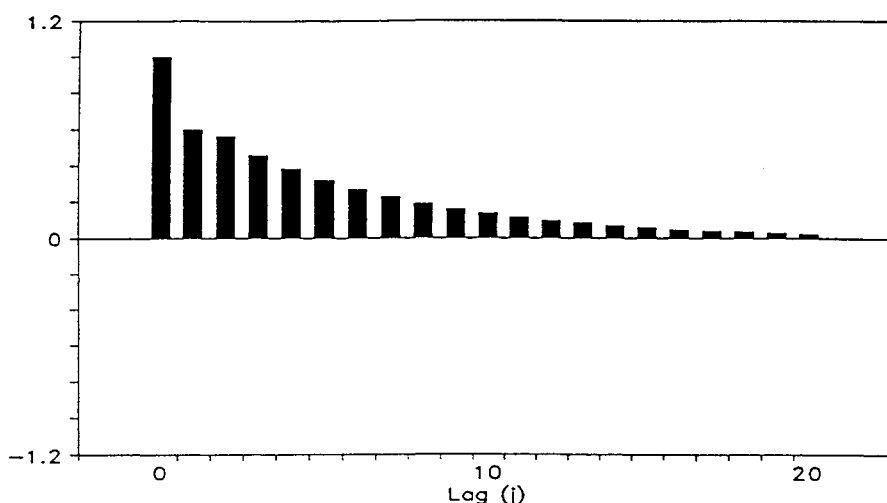
Thus

$$\lambda_2^j = R^j[e^{-i\theta j}] = R^j[\cos(\theta j) - i\sin(\theta j)]. \quad [1.2.38]$$

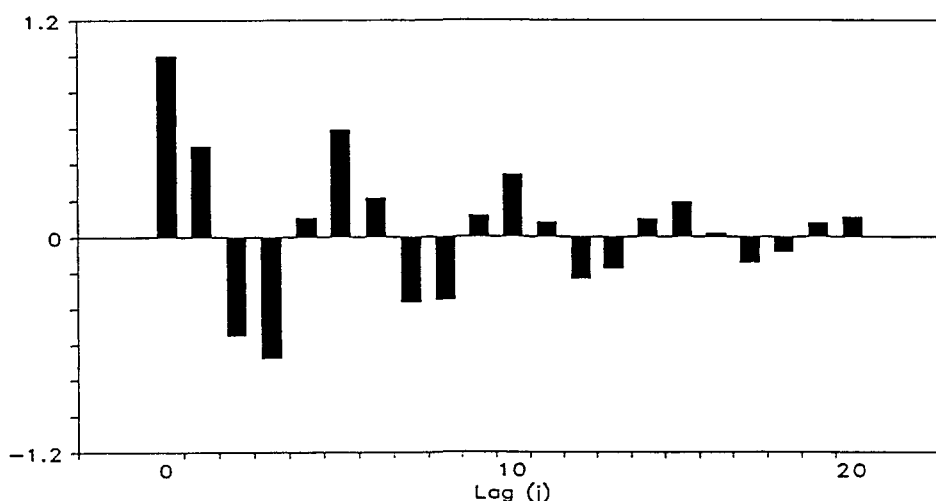
³Again, if one's purpose is solely to generate a numerical plot as in Figure 1.4, the easiest approach is numerical simulation of the system.

⁴See equation [A.3.25] in the Mathematical Review (Appendix A) at the end of the book.

⁵See equation [A.3.26].



(a) $\phi_1 = 0.6, \phi_2 = 0.2$



(b) $\phi_1 = 0.5, \phi_2 = -0.8$

FIGURE 1.4 Dynamic multiplier for second-order difference equation for different values of ϕ_1 and ϕ_2 (plot of $\partial y_{t+j}/\partial w_t$ as a function of the lag j).

Substituting [1.2.37] and [1.2.38] into [1.2.29] gives the contribution of the complex conjugates to the dynamic multiplier $\partial y_{t+j}/\partial w_t$:

$$\begin{aligned} c_1 \lambda_1^j + c_2 \lambda_2^j &= c_1 R^j [\cos(\theta j) + i \sin(\theta j)] + c_2 R^j [\cos(\theta j) - i \sin(\theta j)] \\ &= [c_1 + c_2] \cdot R^j \cos(\theta j) + i [c_1 - c_2] \cdot R^j \sin(\theta j). \end{aligned} \quad [1.2.39]$$

The appearance of the imaginary number i in [1.2.39] may seem a little troubling. After all, this calculation was intended to give the effect of a change in the real-valued variable w_t on the real-valued variable y_{t+j} as predicted by the real-valued system [1.2.1], and it would be odd indeed if the correct answer involved the imaginary number i ! Fortunately, it turns out from [1.2.25] that if λ_1 and λ_2 are complex conjugates, then c_1 and c_2 are complex conjugates; that is, they can

be written as

$$\begin{aligned}c_1 &= \alpha + \beta i \\c_2 &= \alpha - \beta i\end{aligned}$$

for some real numbers α and β . Substituting these expressions into [1.2.39] yields

$$\begin{aligned}c_1\lambda_1^j + c_2\lambda_2^j &= [(\alpha + \beta i) + (\alpha - \beta i)] \cdot R^j \cos(\theta j) + i[(\alpha + \beta i) - (\alpha - \beta i)] \cdot R^j \sin(\theta j) \\&= [2\alpha] \cdot R^j \cos(\theta j) + i[2\beta i] \cdot R^j \sin(\theta j) \\&= 2\alpha R^j \cos(\theta j) - 2\beta R^j \sin(\theta j),\end{aligned}$$

which is strictly real.

Thus, when some of the eigenvalues are complex, they contribute terms proportional to $R^j \cos(\theta j)$ and $R^j \sin(\theta j)$ to the dynamic multiplier $\partial y_{t+j}/\partial w_t$. Note that if $R = 1$ —that is, if the complex eigenvalues have unit modulus—the multipliers are periodic sine and cosine functions of j . A given increase in w_t increases y_{t+j} for some ranges of j and decreases y_{t+j} over other ranges, with the impulse never dying out as $j \rightarrow \infty$. If the complex eigenvalues are less than 1 in modulus ($R < 1$), the impulse again follows a sinusoidal pattern though its amplitude decays at the rate R^j . If the complex eigenvalues are greater than 1 in modulus ($R > 1$), the amplitude of the sinusoids explodes at the rate R^j .

For an example of dynamic behavior characterized by decaying sinusoids, consider the second-order system

$$y_t = 0.5y_{t-1} - 0.8y_{t-2} + w_t.$$

The eigenvalues for this system are given from [1.2.14] and [1.2.15]:

$$\begin{aligned}\lambda_1 &= \frac{0.5 + \sqrt{(0.5)^2 - 4(0.8)}}{2} = 0.25 + 0.86i \\ \lambda_2 &= \frac{0.5 - \sqrt{(0.5)^2 - 4(0.8)}}{2} = 0.25 - 0.86i,\end{aligned}$$

with modulus

$$R = \sqrt{(0.25)^2 + (0.86)^2} = 0.9.$$

Since $R < 1$, the dynamic multiplier follows a pattern of damped oscillation plotted in panel (b) of Figure 1.4. The frequency⁶ of these oscillations is given by the parameter θ in [1.2.39], which was defined implicitly by

$$\cos(\theta) = a/R = (0.25)/(0.9) = 0.28$$

or

$$\theta = 1.29.$$

The cycles associated with the dynamic multiplier function [1.2.39] thus have a period of

$$\frac{2\pi}{\theta} = \frac{(2)(3.14159)}{1.29} = 4.9;$$

that is, the peaks in the pattern in panel (b) of Figure 1.4 appear about five periods apart.

⁶See Section A.1 of the Mathematical Review (Appendix A) at the end of the book for a discussion of the frequency and period of a sinusoidal function.

Solution of a Second-Order Difference Equation with Distinct Eigenvalues

The second-order difference equation ($p = 2$) comes up sufficiently often that it is useful to summarize the properties of the solution as a general function of ϕ_1 and ϕ_2 , which we now do.⁷

The eigenvalues λ_1 and λ_2 in [1.2.14] and [1.2.15] are complex whenever

$$\phi_1^2 + 4\phi_2 < 0,$$

or whenever (ϕ_1, ϕ_2) lies below the parabola indicated in Figure 1.5. For the case of complex eigenvalues, the modulus R satisfies

$$R^2 = a^2 + b^2,$$

or, from [1.2.34] and [1.2.35],

$$R^2 = (\phi_1/2)^2 - (\phi_1^2 + 4\phi_2)/4 = -\phi_2.$$

Thus, a system with complex eigenvalues is explosive whenever $\phi_2 < -1$. Also, when the eigenvalues are complex, the frequency of oscillations is given by

$$\theta = \cos^{-1}(a/R) = \cos^{-1}[\phi_1/(2\sqrt{-\phi_2})],$$

where “ $\cos^{-1}(x)$ ” denotes the inverse of the cosine function, or the radian measure of an angle whose cosine is x .

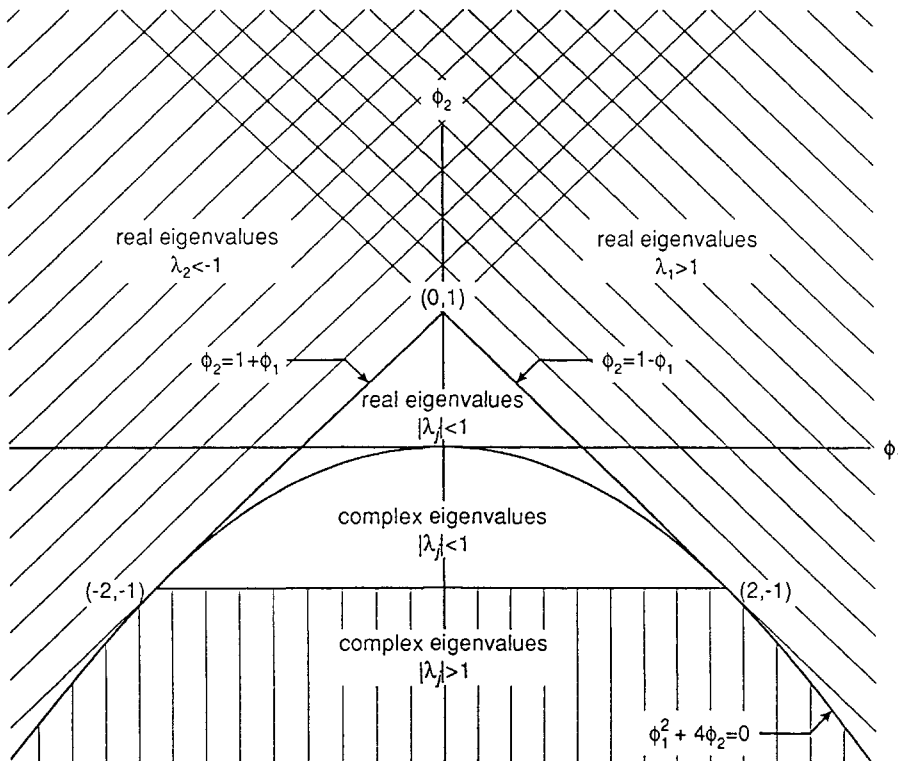


FIGURE 1.5 Summary of dynamics for a second-order difference equation.

⁷This discussion closely follows Sargent (1987, pp. 188–89).

For the case of real eigenvalues, the arithmetically larger eigenvalue (λ_1) will be greater than unity whenever

$$\frac{\phi_1 + \sqrt{\phi_1^2 + 4\phi_2}}{2} > 1$$

or

$$\sqrt{\phi_1^2 + 4\phi_2} > 2 - \phi_1.$$

Assuming that λ_1 is real, the left side of this expression is a positive number and the inequality would be satisfied for any value of $\phi_1 > 2$. If, on the other hand, $\phi_1 < 2$, we can square both sides to conclude that λ_1 will exceed unity whenever

$$\phi_1^2 + 4\phi_2 > 4 - 4\phi_1 + \phi_1^2$$

or

$$\phi_2 > 1 - \phi_1.$$

Thus, in the real region, λ_1 will be greater than unity either if $\phi_1 > 2$ or if (ϕ_1, ϕ_2) lies northeast of the line $\phi_2 = 1 - \phi_1$ in Figure 1.5. Similarly, with real eigenvalues, the arithmetically smaller eigenvalue (λ_2) will be less than -1 whenever

$$\begin{aligned} \frac{\phi_1 - \sqrt{\phi_1^2 + 4\phi_2}}{2} &< -1 \\ -\sqrt{\phi_1^2 + 4\phi_2} &< -2 - \phi_1 \\ \sqrt{\phi_1^2 + 4\phi_2} &> 2 + \phi_1. \end{aligned}$$

Again, if $\phi_1 < -2$, this must be satisfied, and in the case when $\phi_1 > -2$, we can square both sides:

$$\begin{aligned} \phi_1^2 + 4\phi_2 &> 4 + 4\phi_1 + \phi_1^2 \\ \phi_2 &> 1 + \phi_1. \end{aligned}$$

Thus, in the real region, λ_2 will be less than -1 if either $\phi_1 < -2$ or (ϕ_1, ϕ_2) lies to the northwest of the line $\phi_2 = 1 + \phi_1$ in Figure 1.5.

The system is thus stable whenever (ϕ_1, ϕ_2) lies within the triangular region of Figure 1.5.

General Solution of a pth-Order Difference Equation with Repeated Eigenvalues

In the more general case of a difference equation for which $\mathbf{F}$ has repeated eigenvalues and $s < p$ linearly independent eigenvectors, result [1.2.17] is generalized by using the Jordan decomposition,

$$\mathbf{F} = \mathbf{M}\mathbf{J}\mathbf{M}^{-1} \quad [1.2.40]$$

where $\mathbf{M}$ is a $(p \times p)$ matrix and $\mathbf{J}$ takes the form

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_2 & \cdots & \mathbf{0} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{J}_s \end{bmatrix}$$

with

$$\mathbf{J}_i = \begin{bmatrix} \lambda_i & 1 & 0 & \cdots & 0 & 0 \\ 0 & \lambda_i & 1 & \cdots & 0 & 0 \\ 0 & 0 & \lambda_i & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_i & 1 \\ 0 & 0 & 0 & \cdots & 0 & \lambda_i \end{bmatrix} \quad [1.2.41]$$

for λ_i an eigenvalue of $\mathbf{F}$. If [1.2.17] is replaced by [1.2.40], then equation [1.2.19] generalizes to

$$\mathbf{F}^j = \mathbf{M}\mathbf{J}^j\mathbf{M}^{-1} \quad [1.2.42]$$

where

$$\mathbf{J}^j = \begin{bmatrix} \mathbf{J}_1^j & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{J}_2^j & \cdots & \mathbf{0} \\ \vdots & \vdots & \cdots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{J}_s^j \end{bmatrix}.$$

Moreover, from [1.2.41], if $\mathbf{J}_i$ is of dimension $(n_i \times n_i)$, then⁸

$$\mathbf{J}_i^j = \begin{bmatrix} \lambda_i^j & (j)\lambda_i^{j-1} & (j)_2\lambda_i^{j-2} & \cdots & (n_i - j + 1)\lambda_i^{j-n_i+1} \\ 0 & \lambda_i^j & (j)\lambda_i^{j-1} & \cdots & (n_i - j)\lambda_i^{j-n_i+2} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_i^j \end{bmatrix} \quad [1.2.43]$$

where

$$\binom{j}{n} \equiv \begin{cases} \frac{j(j-1)(j-2)\cdots(j-n+1)}{n(n-1)\cdots 3 \cdot 2 \cdot 1} & \text{for } j \geq n \\ 0 & \text{otherwise.} \end{cases}$$

Equation [1.2.43] may be verified by induction by multiplying [1.2.41] by [1.2.43] and noticing that $\binom{j}{n} + \binom{j-1}{n-1} = \binom{j}{n-1}$.

For example, consider again the second-order difference equation, this time with repeated roots. Then

$$\mathbf{F}^j = \mathbf{M} \begin{bmatrix} \lambda^j & j\lambda^{j-1} \\ 0 & \lambda^j \end{bmatrix} \mathbf{M}^{-1},$$

so that the dynamic multiplier takes the form

$$\frac{\partial y_{t+j}}{\partial w_t} = f_{11}^{(j)} = k_1 \lambda^j + k_2 j \lambda^{j-1}.$$

Long-Run and Present-Value Calculations

If the eigenvalues are all less than 1 in modulus, then $\mathbf{F}^j$ in [1.2.9] goes to zero as j becomes large. If all values of w and y are taken to be bounded, we can

⁸This expression is taken from Chiang (1980, p. 444).

think of a “solution” of y_t in terms of the infinite history of w ,

$$y_t = w_t + \psi_1 w_{t-1} + \psi_2 w_{t-2} + \psi_3 w_{t-3} + \cdots, \quad [1.2.44]$$

where ψ_j is given by the $(1, 1)$ element of $\mathbf{F}^j$ and takes the particular form of [1.2.29] in the case of distinct eigenvalues.

It is also straightforward to calculate the effect on the present value of y of a transitory increase in w . This is simplest to find if we first consider the slightly more general problem of the hypothetical consequences of a change in any element of the vector $\mathbf{v}_t$ on any element of ξ_{t+j} in a general system of the form of [1.2.5]. The answer to this more general problem can be inferred immediately from [1.2.9]:

$$\frac{\partial \xi_{t+j}}{\partial \mathbf{v}_t'} = \mathbf{F}^j. \quad [1.2.45]$$

The true dynamic multiplier of interest, $\partial y_{t+j}/\partial w_t$, is just the $(1, 1)$ element of the $(p \times p)$ matrix in [1.2.45]. The effect on the present value of ξ of a change in $\mathbf{v}$ is given by

$$\frac{\partial \sum_{j=0}^{\infty} \beta^j \xi_{t+j}}{\partial \mathbf{v}_t'} = \sum_{j=0}^{\infty} \beta^j \mathbf{F}^j = (\mathbf{I}_p - \beta \mathbf{F})^{-1}, \quad [1.2.46]$$

provided that the eigenvalues of $\mathbf{F}$ are all less than β^{-1} in modulus. The effect on the present value of y of a change in w ,

$$\frac{\partial \sum_{j=0}^{\infty} \beta^j y_{t+j}}{\partial w_t},$$

is thus the $(1, 1)$ element of the $(p \times p)$ matrix in [1.2.46]. This value is given by the following proposition.

Proposition 1.3: *If the eigenvalues of the $(p \times p)$ matrix $\mathbf{F}$ defined in [1.2.3] are all less than β^{-1} in modulus, then the matrix $(\mathbf{I}_p - \beta \mathbf{F})^{-1}$ exists and the effect of w on the present value of y is given by its $(1, 1)$ element:*

$$1/(1 - \phi_1 \beta - \phi_2 \beta^2 - \cdots - \phi_{p-1} \beta^{p-1} - \phi_p \beta^p).$$

Note that Proposition 1.3 includes the earlier result for a first-order system (equation [1.1.14]) as a special case.

The cumulative effect of a one-time change in w_t on $y_t, y_{t+1}, \dots$ can be considered a special case of Proposition 1.3 with no discounting. Setting $\beta = 1$ in Proposition 1.3 shows that, provided the eigenvalues of $\mathbf{F}$ are all less than 1 in modulus, the cumulative effect of a one-time change in w on y is given by

$$\sum_{j=0}^{\infty} \frac{\partial y_{t+j}}{\partial w_t} = 1/(1 - \phi_1 - \phi_2 - \cdots - \phi_p). \quad [1.2.47]$$

Notice again that [1.2.47] can alternatively be interpreted as giving the eventual long-run effect on y of a permanent change in w :

$$\lim_{j \rightarrow \infty} \frac{\partial y_{t+j}}{\partial w_t} + \frac{\partial y_{t+j}}{\partial w_{t+1}} + \frac{\partial y_{t+j}}{\partial w_{t+2}} + \cdots + \frac{\partial y_{t+j}}{\partial w_{t+j}} = 1/(1 - \phi_1 - \phi_2 - \cdots - \phi_p).$$

APPENDIX 1.A. Proofs of Chapter 1 Propositions

■ **Proof of Proposition 1.1.** The eigenvalues of $\mathbf{F}$ satisfy

$$|\mathbf{F} - \lambda \mathbf{I}_p| = 0. \quad [1.A.1]$$

For the matrix $\mathbf{F}$ defined in equation [1.2.3], this determinant would be

$$\begin{vmatrix} \phi_1 & \phi_2 & \phi_3 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{vmatrix} = \begin{vmatrix} \lambda & 0 & 0 & \cdots & 0 & 0 \\ 0 & \lambda & 0 & \cdots & 0 & 0 \\ 0 & 0 & \lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & \lambda \end{vmatrix} \\ = \begin{vmatrix} (\phi_1 - \lambda) & \phi_2 & \phi_3 & \cdots & \phi_{p-1} & \phi_p \\ 1 & -\lambda & 0 & \cdots & 0 & 0 \\ 0 & 1 & -\lambda & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -\lambda \end{vmatrix}. \quad [1.A.2]$$

Recall that if we multiply a column of a matrix by a constant and add the result to another column, the determinant of the matrix is unchanged. If we multiply the p th column of the matrix in [1.A.2] by $(1/\lambda)$ and add the result to the $(p-1)$ th column, the result is a matrix with the same determinant as that in [1.A.2]:

$$|\mathbf{F} - \lambda \mathbf{I}_p| = \begin{vmatrix} \phi_1 - \lambda & \phi_2 & \phi_3 & \cdots & \phi_{p-2} & \phi_{p-1} + (\phi_p/\lambda) & \phi_p \\ 1 & -\lambda & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & -\lambda & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -\lambda & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & -\lambda \end{vmatrix}.$$

Next, multiply the $(p-1)$ th column by $(1/\lambda)$ and add the result to the $(p-2)$ th column:

$$|\mathbf{F} - \lambda \mathbf{I}_p| = \begin{vmatrix} \phi_1 - \lambda & \phi_2 & \phi_3 & \cdots & \phi_{p-2} + \phi_{p-1}/\lambda + \phi_p/\lambda^2 & \phi_{p-1} + \phi_p/\lambda & \phi_p \\ 1 & -\lambda & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & -\lambda & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & -\lambda & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & -\lambda \end{vmatrix}.$$

Continuing in this fashion shows [1.A.1] to be equivalent to the determinant of the following upper triangular matrix:

$$|\mathbf{F} - \lambda \mathbf{I}_p| = \begin{vmatrix} \phi_1 - \lambda + \phi_2/\lambda + \phi_3/\lambda^2 + \cdots + \phi_p/\lambda^{p-1} & \phi_2 + \phi_3/\lambda + \phi_4/\lambda^2 + \cdots + \phi_p/\lambda^{p-2} & \cdots & \phi_{p-1} + \phi_p/\lambda & \phi_p \\ 0 & -\lambda & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & -\lambda & 0 \\ 0 & 0 & \cdots & 0 & -\lambda \end{vmatrix}.$$

But the determinant of an upper triangular matrix is simply the product of the terms along the principal diagonal:

$$\begin{aligned} |\mathbf{F} - \lambda \mathbf{I}_p| &= [\phi_1 - \lambda + \phi_2/\lambda + \phi_3/\lambda^2 + \cdots + \phi_p/\lambda^{p-1}] \cdot [-\lambda]^{p-1} \\ &= (-1)^p \cdot [\lambda^p - \phi_1 \lambda^{p-1} - \phi_2 \lambda^{p-2} - \cdots - \phi_p]. \end{aligned} \quad [1.A.3]$$

The eigenvalues of $\mathbf{F}$ are thus the values of λ for which [1.A.3] is zero, or for which

$$\lambda^p - \phi_1 \lambda^{p-1} - \phi_2 \lambda^{p-2} - \cdots - \phi_p = 0,$$

as asserted in Proposition 1.1. ■

■ **Proof of Proposition 1.2.** Assuming that the eigenvalues $(\lambda_1, \lambda_2, \dots, \lambda_p)$ are distinct, the matrix $\mathbf{T}$ in equation [1.2.17] can be constructed from the eigenvectors of $\mathbf{F}$. Let $\mathbf{t}_i$ denote the following $(p \times 1)$ vector,

$$\mathbf{t}_i = \begin{bmatrix} \lambda_i^{p-1} \\ \lambda_i^{p-2} \\ \lambda_i^{p-3} \\ \vdots \\ \lambda_i^1 \\ 1 \end{bmatrix}, \quad [1.A.4]$$

where λ_i denotes the i th eigenvalue of $\mathbf{F}$. Notice

$$\begin{aligned} \mathbf{F}\mathbf{t}_i &= \begin{bmatrix} \phi_1 & \phi_2 & \phi_3 & \cdots & \phi_{p-1} & \phi_p \\ 1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \end{bmatrix} \begin{bmatrix} \lambda_i^{p-1} \\ \lambda_i^{p-2} \\ \lambda_i^{p-3} \\ \vdots \\ \lambda_i^1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} \phi_1 \lambda_i^{p-1} + \phi_2 \lambda_i^{p-2} + \phi_3 \lambda_i^{p-3} + \cdots + \phi_{p-1} \lambda_i + \phi_p \\ \lambda_i^{p-1} \\ \lambda_i^{p-2} \\ \vdots \\ \lambda_i^2 \\ \lambda_i \end{bmatrix}. \end{aligned} \quad [1.A.5]$$

Since λ_i is an eigenvalue of $\mathbf{F}$, it satisfies [1.2.16]:

$$\lambda_i^p - \phi_1 \lambda_i^{p-1} - \phi_2 \lambda_i^{p-2} - \cdots - \phi_{p-1} \lambda_i - \phi_p = 0. \quad [1.A.6]$$

Substituting [1.A.6] into [1.A.5] reveals

$$\mathbf{F}\mathbf{t}_i = \begin{bmatrix} \lambda_i^p \\ \lambda_i^{p-1} \\ \lambda_i^{p-2} \\ \vdots \\ \lambda_i^2 \\ \lambda_i \end{bmatrix} = \lambda_i \begin{bmatrix} \lambda_i^{p-1} \\ \lambda_i^{p-2} \\ \lambda_i^{p-3} \\ \vdots \\ \lambda_i^1 \\ 1 \end{bmatrix}$$

or

$$\mathbf{F}\mathbf{t}_i = \lambda_i \mathbf{t}_i. \quad [1.A.7]$$

Thus $\mathbf{t}_i$ is an eigenvector of $\mathbf{F}$ associated with the eigenvalue λ_i .

We can calculate the matrix $\mathbf{T}$ by combining the eigenvectors $(\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_p)$ into a $(p \times p)$ matrix

$$\mathbf{T} = [\mathbf{t}_1 \quad \mathbf{t}_2 \quad \cdots \quad \mathbf{t}_p]. \quad [1.A.8]$$

To calculate the particular values for c_i in equation [1.2.21], recall that $\mathbf{T}^{-1}$ is characterized by

$$\mathbf{T}\mathbf{T}^{-1} = \mathbf{I}_p, \quad [1.A.9]$$

where $\mathbf{T}$ is given by [1.A.4] and [1.A.8]. Writing out the first column of the matrix system of equations [1.A.9] explicitly, we have

$$\begin{bmatrix} \lambda_1^{p-1} & \lambda_2^{p-1} & \cdots & \lambda_p^{p-1} \\ \lambda_1^{p-2} & \lambda_2^{p-2} & \cdots & \lambda_p^{p-2} \\ \lambda_1^{p-3} & \lambda_2^{p-3} & \cdots & \lambda_p^{p-3} \\ \vdots & \vdots & \cdots & \vdots \\ \lambda_1^1 & \lambda_2^1 & \cdots & \lambda_p^1 \\ 1 & 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} t^{11} \\ t^{21} \\ t^{31} \\ \vdots \\ t^{p-1,1} \\ t^{p1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}.$$

This gives a system of p linear equations in the p unknowns $(t^{11}, t^{21}, \dots, t^{p1})$. Provided that the λ_i are all distinct, the solution can be shown to be⁹

$$\begin{aligned} t^{11} &= \frac{1}{(\lambda_1 - \lambda_2)(\lambda_1 - \lambda_3) \cdots (\lambda_1 - \lambda_p)} \\ t^{21} &= \frac{1}{(\lambda_2 - \lambda_1)(\lambda_2 - \lambda_3) \cdots (\lambda_2 - \lambda_p)} \\ &\vdots \\ t^{p1} &= \frac{1}{(\lambda_p - \lambda_1)(\lambda_p - \lambda_2) \cdots (\lambda_p - \lambda_{p-1})}. \end{aligned}$$

Substituting these values into [1.2.21] gives equation [1.2.25]. ■

■ **Proof of Proposition 1.3.** The first claim in this proposition is that if the eigenvalues of $\mathbf{F}$ are less than β^{-1} in modulus, then the inverse of $(\mathbf{I}_p - \beta\mathbf{F})$ exists. Suppose the inverse of $(\mathbf{I}_p - \beta\mathbf{F})$ did not exist. Then the determinant $|\mathbf{I}_p - \beta\mathbf{F}|$ would have to be zero. But

$$|\mathbf{I}_p - \beta\mathbf{F}| = |-\beta \cdot [\mathbf{F} - \beta^{-1}\mathbf{I}_p]| = (-\beta)^p |\mathbf{F} - \beta^{-1}\mathbf{I}_p|,$$

so that $|\mathbf{F} - \beta^{-1}\mathbf{I}_p|$ would have to be zero whenever the inverse of $(\mathbf{I}_p - \beta\mathbf{F})$ fails to exist. But this would mean that β^{-1} is an eigenvalue of $\mathbf{F}$, which is ruled out by the assumption that all eigenvalues of $\mathbf{F}$ are strictly less than β^{-1} in modulus. Thus, the matrix $\mathbf{I}_p - \beta\mathbf{F}$ must be nonsingular.

Since $(\mathbf{I}_p - \beta\mathbf{F})^{-1}$ exists, it satisfies the equation

$$(\mathbf{I}_p - \beta\mathbf{F})^{-1}(\mathbf{I}_p - \beta\mathbf{F}) = \mathbf{I}_p. \quad [1.A.10]$$

Let x_{ij} denote the row i , column j element of $(\mathbf{I}_p - \beta\mathbf{F})^{-1}$, and write [1.A.10] as

$$\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \cdots & \vdots \\ x_{p1} & x_{p2} & \cdots & x_{pp} \end{bmatrix} \begin{bmatrix} 1 - \beta\phi_1 & -\beta\phi_2 & \cdots & -\beta\phi_{p-1} & -\beta\phi_p \\ -\beta & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & -\beta & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}. \quad [1.A.11]$$

The task is then to find the $(1, 1)$ element of $(\mathbf{I}_p - \beta\mathbf{F})^{-1}$, that is, to find the value of x_{11} . To do this we need only consider the first row of equations in [1.A.11]:

$$[x_{11} \ x_{12} \ \cdots \ x_{1p}] \begin{bmatrix} 1 - \beta\phi_1 & -\beta\phi_2 & \cdots & -\beta\phi_{p-1} & -\beta\phi_p \\ -\beta & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & -\beta & 1 \end{bmatrix} = [1 \ 0 \ \cdots \ 0 \ 0]. \quad [1.A.12]$$

⁹See Lemma 2 of Chiang (1980, p. 144).

Consider postmultiplying this system of equations by a matrix with 1s along the principal diagonal, β in the row p , column $p - 1$ position, and 0s elsewhere:

$$\begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & \beta & 1 \end{bmatrix}.$$

The effect of this operation is to multiply the p th column of a matrix by β and add the result to the $(p - 1)$ th column:

$$[x_{11} \ x_{12} \ \cdots \ x_{1p}] \begin{bmatrix} 1 - \beta\phi_1 & -\beta\phi_2 & \cdots & -\beta\phi_{p-1} - \beta^2\phi_p & -\beta\phi_p \\ -\beta & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} = [1 \ 0 \ \cdots \ 0 \ 0].$$

Next multiply the $(p - 1)$ th column by β and add the result to the $(p - 2)$ th column. Proceeding in this fashion, we arrive at

$$[x_{11} \ x_{12} \ \cdots \ x_{1p}] \times \begin{bmatrix} 1 - \beta\phi_1 - \beta^2\phi_2 - \cdots - \beta^{p-1}\phi_{p-1} - \beta^p\phi_p & -\beta\phi_2 - \beta^2\phi_3 - \cdots - \beta^{p-1}\phi_p & \cdots & -\beta\phi_{p-1} - \beta^2\phi_p & -\beta\phi_p \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix} = [1 \ 0 \ \cdots \ 0 \ 0]. \quad [1.A.13]$$

The first equation in [1.A.13] states that

$$x_{11} \cdot (1 - \beta\phi_1 - \beta^2\phi_2 - \cdots - \beta^{p-1}\phi_{p-1} - \beta^p\phi_p) = 1$$

or

$$x_{11} = 1/(1 - \beta\phi_1 - \beta^2\phi_2 - \cdots - \beta^p\phi_p),$$

as claimed in Proposition 1.3. ■

Chapter 1 References

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